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Final Report for Holiday Inn

As part of the Holiday Inn Data Science Team, I have been tasked with identifying bookings with our hotels that have a high risk of being cancelled. The risk of cancellation will be presented using values between 0 and 1 as a representation of the probability of a cancellation. The goal of this project is to identify bookings with a high risk of being cancelled, and to prevent those potential cancellations using specific strategies to retain our bookings. I work closely with the hotel management team to implement actionable strategies to stop cancellations and boost booking retention. We target these high-risk bookings with special offers and promotions. In terms of a supervised classification problem to address the business need, the goal is to correctly classify whether a booking will be cancelled or not using the booking data we have recorded previously, enabling us to make predictions utilizing this particular data. We leverage information regarding pricing, bookings, and guests to estimate the risk of cancellation using the trained classification model.

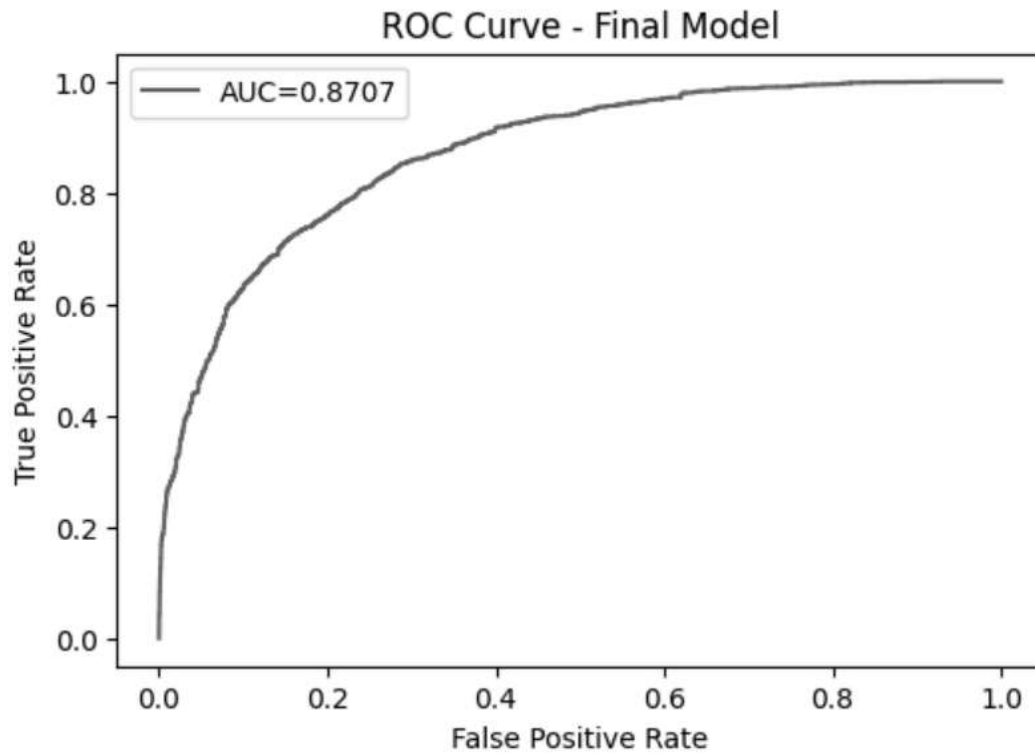
For the first aspect of the preliminary results, I began by addressing the data preprocessing and feature engineering needs that I outlined in the Analytic Plan. After loading the necessary libraries for conducting this stage of the project, I then loaded the project dataset. The first matter of business was dropping the 'Booking_ID' column because 'Booking_ID' serves as a unique identifier and does not give us much predictive power for predicting the booking status. The next step was to one-hot encode the target variable 'booking_status' into binary form (0 or 1) for classification. Then, I extracted some features from 'arrival_date' to split the variable into numerous variables for different purposes, like looking at the arrival day, week, weekend versus weekday, week of the year, seasonality, and reference date (to be able to identify

trends in cancellations). In addition to this feature engineering, I created variables to measure the length of a booking as well as a cancellation ratio. Additionally, I one-hot encoded all of the remaining categorical variables: 'type_of_meal_plan', 'room_type_reserved', 'market_segment_type', and 'season'. I standardized the numeric features in the model through the use of scaling, working to improve the training stability of the neural network. Following the scaling process, I then split the data into train and test sets, where I used stratification to address potential issues of class imbalance.

I built a simple feedforward neural network model where I used 1 hidden layer, eventually using neuron sizes of 32, 64, and 128 after tuning the hyperparameters. I also used learning rates of 0.001 and 0.0005, as well as epoch levels of 50 and 100. For an activation function, I used 'ReLU()' to introduce non-linearity, while using 'Sigmoid()' to output probabilities between 0 and 1 for binary classification purposes. The Binary Cross Entropy Loss function was utilized here since I was working with binary classification. Additionally, I used 'Adam()' in an effort to lead to reliable and optimal results for the neural network.

I built 12 models using the defined sets of variable hyperparameters, using several loops, to gauge how certain parameters might affect the model. The best, most accurate model had an accuracy of 81.21%, using 128 hidden neurons, a learning rate of 0.001, 100 epochs, and returning an AUC of 87.07%, which I graphed to display the visual results.

Final model -> Accuracy: 0.8121, AUC: 0.8707



From our results, we can conclude that increasing the size of the hidden layer and the number of epochs improved the performance on average. Our learning curves suggest that the model is not overfitting significantly due to the fact that the validation performance improves as the number of layers and epochs increases, seeming to get stronger over time as the model learns and grows, learning patterns within the data fairly well.

```
# save model results to a dataframe
results_df = pd.DataFrame(results, columns=['Hidden', 'LR', 'Epochs', 'Accuracy', 'AUC'])
results_df = results_df.sort_values(by='AUC', ascending=False)
print(results_df)
```

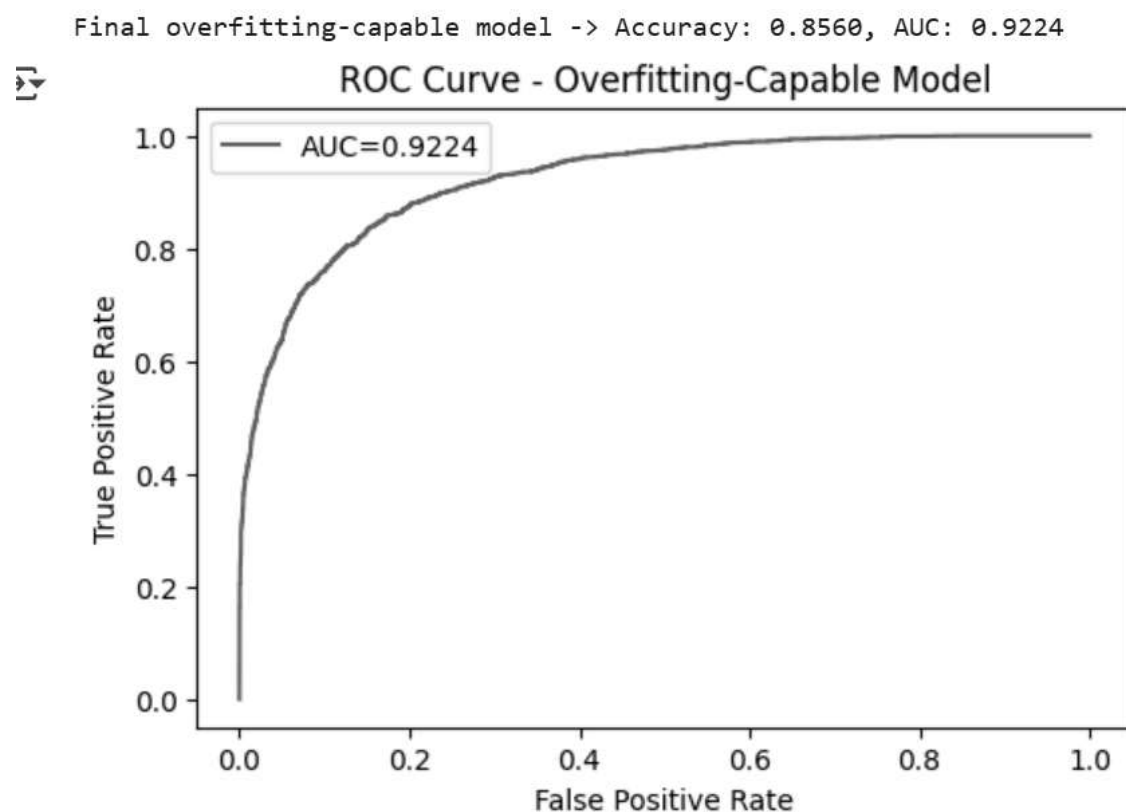
	Hidden	LR	Epochs	Accuracy	AUC
9	128	0.0010	100	0.808637	0.874124
5	64	0.0010	100	0.785458	0.844059
11	128	0.0005	100	0.767522	0.827947
8	128	0.0010	50	0.759244	0.827483
1	32	0.0010	100	0.770833	0.825016
7	64	0.0005	100	0.744895	0.814775
3	32	0.0005	100	0.722268	0.801119
4	64	0.0010	50	0.733030	0.800752
6	64	0.0005	50	0.690811	0.757800
2	32	0.0005	50	0.728753	0.757627
0	32	0.0010	50	0.713024	0.756457
10	128	0.0005	50	0.691777	0.712765

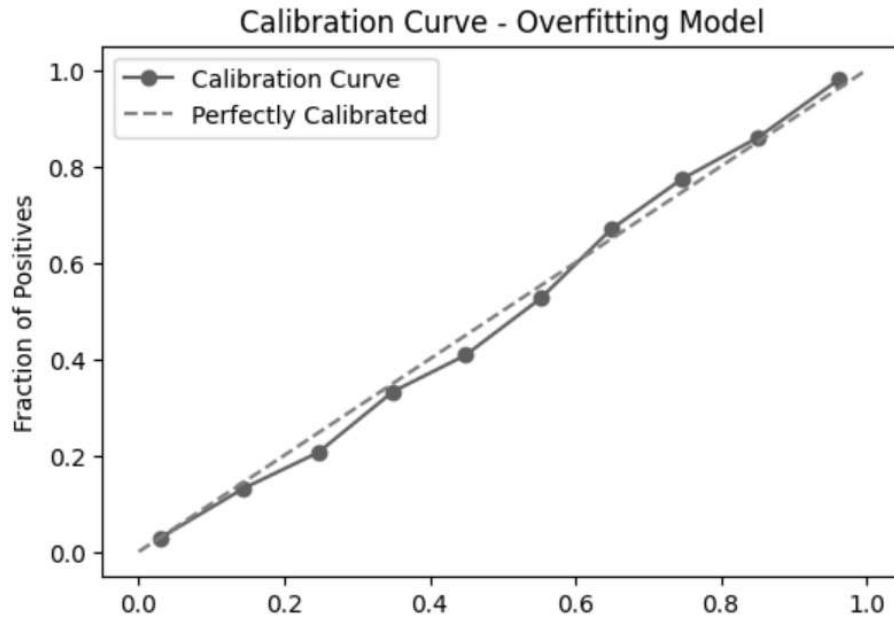
I also took a quick look at the correlation between the target variable and the predictors, where lead_time was the most closely correlated variable to booking status. Many of the predictors demonstrated weak relationships to booking status, making it difficult to put a pin in the most impactful factors in a booking cancellation.

booking_status	1.000000
lead_time	0.438629
arrival_year	0.179762
days_since_reference	0.172698
avg_price_per_room	0.142721
season_summer	0.110196
market_segment_type_online	0.106131
total_nights	0.103474
no_of_week_nights	0.092904
no_of_adults	0.086671
type_of_meal_plan_meal_plan_2	0.086550
no_of_weekend_nights	0.061704
no_of_children	0.033033
room_type_reserved_room_type6	0.032888
arrival_dayofweek	0.030317
is_weekend	0.029501
season_spring	0.017502
room_type_reserved_room_type4	0.013443
arrival_day	0.011109
type_of_meal_plan_not_selected	0.002838
room_type_reserved_room_type2	0.000506
room_type_reserved_room_type3	-0.001246
type_of_meal_plan_meal_plan_3	-0.003198
room_type_reserved_room_type5	-0.009839

Within the final report stage of the project, I created a new set of feedforward neural network models that are capable of overfitting, utilizing suggestions made in my preliminary results and the feedback I received regarding feature engineering. In addition to the creation of the new set of models, I added a calibration curve for the best original final model and for the best new final model, to demonstrate and compare the results for both models. Similarly to the original models I created, I tested different combinations of each feature to find the best combination. For the new final model itself, I used hidden neuron levels of 128 and 256 due to the results of the original results being better for the higher the level of neurons, as well as using two hidden layers. I kept the learning the rates the same to make comparison a fair scale. I increased the epoch level to 100 and 200 given the previous results being better for the epoch

levels that were higher (100). Then as suggested in the preliminary results feedback, I implemented dropout rates of 0.2 and 0.5 to control overfitting, ensuring that certain neurons aren't too heavily relied upon, creating a better balance and improved validity/accuracy to my results. The best new final model created was where the hidden neurons were a value of 256, epoch level of 200, learning rate of 0.001, and a dropout rate of 0.2. This model had an accuracy of 85.6% with an AUC of 92.24%. We can also see in the calibration curve for the best new final model the curve is well calibrated as it closely follows the diagonal line, whereas the best original final model displays similar results but not quite as strong as the new model's results, indicating that the new model is stronger and more accurate for that matter.





Through the creating of the setup of this new modeling, we succeeded in building a stronger, more accurate model in predicting our ultimate goal of identifying hotel booking cancellations. This information is powerful because with this being the best model, we can put our trust in the model and its accuracy in making predictions that we can act on for business purposes. We can prioritize bookings that are at risk and roll out offers in a timely manner in an effort to prevent the cancellation. Going forward, we could continue to play around with the features of our model to try to build an even stronger and more accurate model, but right now our final model already appears to be quite successful. We can utilize our model for prediction making purposes, where we use our probability outputs to encourage targeted promotional offers and prioritize customers/bookings that are at risk for cancellation. For the future, when customers cancel their booking with us, we can implement a requirement for the customer to explain to us why they canceled their booking. We can do this best using a survey where we ask them questions about their experience and the reason that most aligns with their reason for

cancelling their booking. We can include a text box for customers to input in a certain number of characters their reason in addition to the questions and answer options that we come up with. Additionally, in the future we can maybe utilize features in our modeling that involve overall customer engagement and perhaps encourage a loyalty program to entice customers to keep their bookings; maybe even implement a penalty policy for loyalty program members to improve booking numbers. If we used a tier system for the program, if loyalty members cancel their booking then we could demote the customer's tier in the system for cancelling their booking (with certain exceptions).